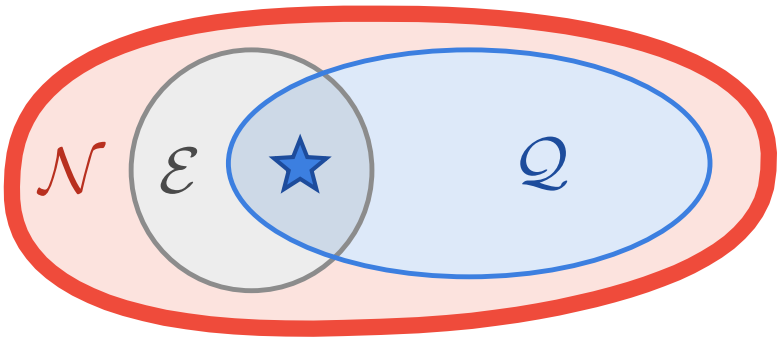
$\mathcal{N}$ quantum systems with depolarizing noise and entropy budget $k$

$\mathcal{Q}$ quantum reservoirs $\mathcal{E}$ measurement-free error correction

**a**

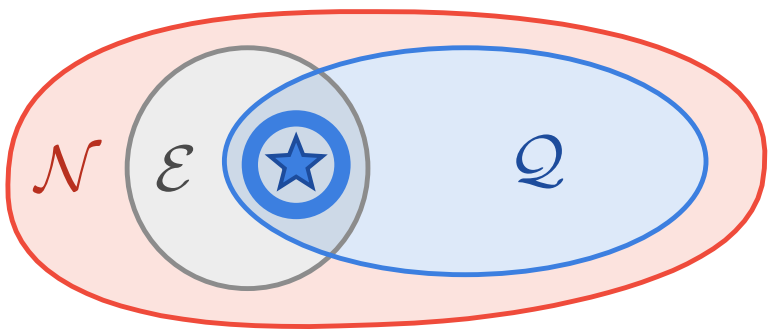



Upper bound

$$\max_{\mathcal{N}} W \leq e^{O(k)}$$

holds for every system in $\mathcal{N}$

**b**

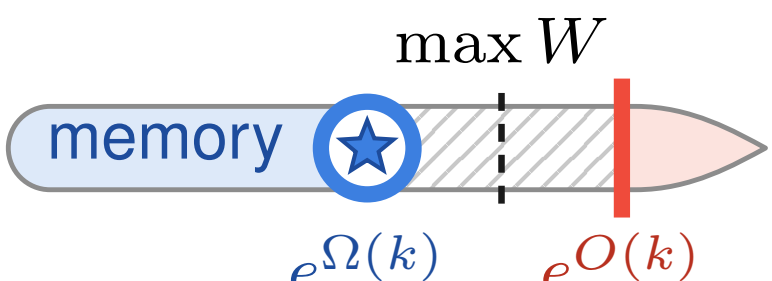



Lower bound

$$\max_{\mathcal{Q}} W \geq e^{\Omega(k)}$$

reached by one reservoir (  )

**c**




Together: optimal scaling

$$e^{\Omega(k)} \leq \max_{\mathcal{Q}} W \leq \max_{\mathcal{N}} W \leq e^{O(k)}$$

both maxima grow as $e^{\Theta(k)}$